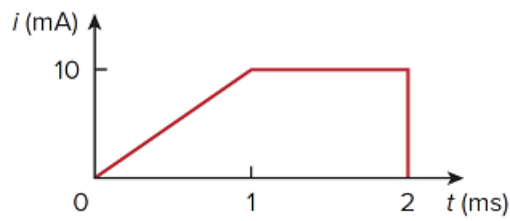


Problem

The current flowing past a point in a device is shown in Fig. 1.25. Calculate the total charge through the point.

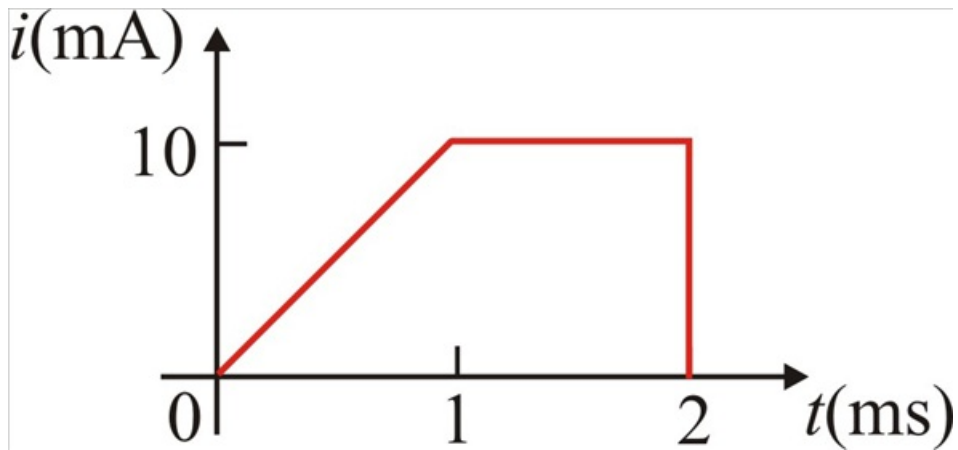
Figure 1.25



Step-by-step solution

Step 1 of 3

Given current waveform



Step 2 of 3

From the above waveform we can define

$$i(t) = \begin{cases} 10t & \text{for } 0 < t < 1 \\ 10 \times 10^{-3} & \text{for } 1 < t < 2 \end{cases}$$

$$q(t) = \int i(t) dt$$

For $0 < t < 1$

$$q(t) = \int_0^{1\text{m}} 10t dt$$

$$q(t) = \left[\frac{10t^2}{2} \right]_0^{1\text{m}}$$

$$q(t) = \frac{10(1 \times 10^{-3})^2}{2} - \frac{10(0)^2}{2}$$

$$q(t) = 5 \times 10^{-6}$$

$$q(t) = 5 \mu\text{C}$$

Step 3 of 3

For $1 < t < 2$

$$q(t) = \int_{1m}^{2m} 10 \times 10^{-3} dt$$

$$q(t) = 10 \times 10^{-3} [t]_{1m}^{2m}$$

$$q(t) = 10 \times 10^{-3} [2 \times 10^{-3} - 1 \times 10^{-3}]$$

$$q(t) = 10 \times 10^{-6}$$

$$q(t) = 10 \mu C$$

Therefore total charge through the point

$$q = 5 \mu C + 10 \mu C$$

$$\boxed{q = 15 \mu C}$$